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Magma flow directions of post Jotnian dolerite sills in central-east Sweden: a magnetic fabric and paleomagnetic survey

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Abstract: An anisotropy of magnetic susceptibility (AMS) and paleomagnetic study has been performed on post-Jotnian basic sills of the Ulvö complex, of the Central Scandinavian Dolerite Group (CSDG). The CSDG intruded some 1.27 Ga ago in the central part of the Fennoscandian Shield and extends over an area of c. 150 000 km^2. Previously presented AMS data of the CSDG sills from south-central and northern Sweden indicated a NW or SE directed magma flow. In this study new measurements of the AMS reveal a fairly uniform pattern of NW–SE lineation directions also for the dolerites of the Ulvö complex. While the magma is flowing in a dyke or a sill, elongated particles become imbricated against the chilled margins, and the AMS can be used to determine not only the lineation but also the true direction of the magma flow. An imbrication pattern defined for one of the sills suggests that the magma propagated from the present NW towards SE. The established paleomagnetic pole position (Plat=–3° and Plon=157°) is in very good agreement with other paleomagnetic data from the Ulvö complex and from other parts of the CSDG. The new AMS data from the Ulvö complex presented in this study fit well in to the magnetic fabric pattern previously presented for the CSDG sills and support a plate tectonic model suggesting that the CSDG is a related to the break up of Baltica from Laurentia at c. 1.27 Ga ago.

Keywords: Anisotropy of magnetic susceptibility (AMS), Paleomagnetism, Dolerite sills, Baltica, Magma flow.

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Introduction

During the Precambrian various generations of dyke complexes intruded in the Fennoscandian Shield. At c. 1.27 Ga dolerite sills and dykes (Welin & Lundqvist 1975; Patchett 1978; Suominen 1991), the Central Scandinavian Dolerite Group (CSDG; Gorbatschev et al. 1979), intruded in the central-northern part of Sweden and south-western part of Finland, extending over an area of c. 150 000 km^2 (Elming & Mattsson 2001, Fig. 1). This group of basic intrusions is coeval with large igneous events in the North Atlantic area, e.g. the Gardar dykes in Greenland and the Mackenzie and Sudbury dykes in Canada.

Previously presented AMS data, mainly from the south central (Jämtland), northern (Västerbotten) and eastern (Satakunta) parts of the CSDG complex (Elming & Mattsson 2001) show a magnetic fabric that indicates a fairly uniform c. NW or SE directed magma flow of the dolerites in Sweden. Elming & Mattsson (2001) interpreted the flow as a result of a major magma intrusion during the break up of Laurentia from Baltica some 1.27 Ga ago.

In this study new AMS and paleomagnetic data are presented from post-Jotnian dolerites outcropping between the cities Örnsköldsvik and Sundsvall along the east coast of central Sweden (e.g. Lundqvist et al. 1990). The purpose of the study is to test if the AMS of the dolerites in this area can be correlated to the NW–SE directed fabrics of the Jämtland and Västerbotten sills as shown by Elming & Mattsson (2001). We will also test the proposed flow direction from the present NW to the SE suggested by Elming & Mattsson (2001). Special efforts were put in collecting samples within a few centimetres of the chilled margins of the sills. While the magma is flowing in the sill, elongated particles become imbricated against the chilled margins, and not only the lineation but also the true direction of the magma flow can be determined (Knight & Walker 1988).

General geology

The Fennoscandian Shield is divided into a number of tectono-magmatic blocks (Fig. 1, inset). These blocks represent a geochronological zonation and an accretionary growth of Fennoscandia during several successive orogenic events, i.e. the Loopian (2.9–2.6 Ga), the Svecofennian (1.9–1.85 Ga), the

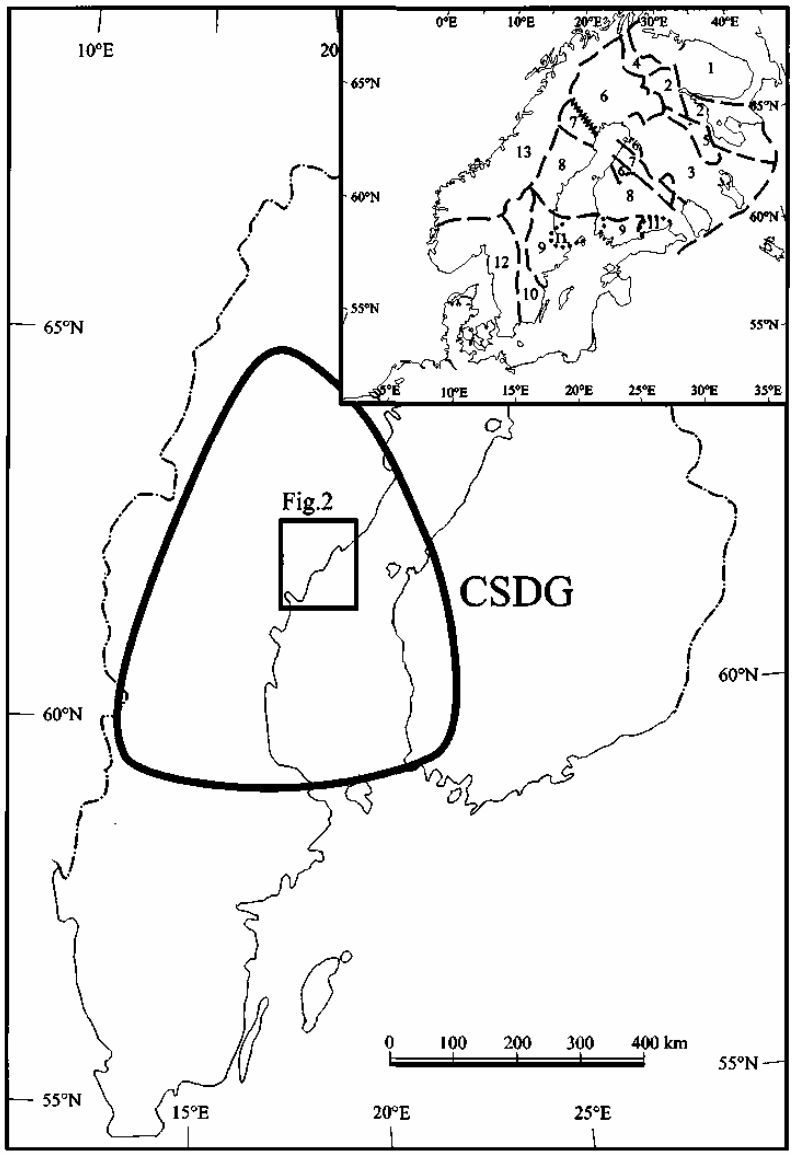


Fig. 1. Generalised geographical distribution of the Central Scandinavian Dolerite Group (CSDG). The square shows the location of the study area of this investigation (see Fig. 2). In the inset the tectonomagmatic block division of the Fennoscandian Shield is shown (modified after Gaal & Gorbatschev 1987; Elming et al. 1993). Archaean province: 1 = Kola, 2 = Belomoria, 3 = Karelia subprovinces. Svecofennian province; 4 = Lappland granulite belt, 5 = Jatulian/ Svecofennian outlier in Karelia, 6 = Northern Svecofennian subprovince, 7 = Skellefte and Raahe Ladoga belts, 8 = Central Svecofennian subprovince, 9 = Southern Svecofennian subprovince, 10 = Transscandinavian Igneous Belt (TIB), 11 = Rapakivi granites, 12 = Southwestern Scandinavian province, 13 = Caledonides.

Gothian, and the Transscandinavian Igneous Belt (TIB; 1.75–1.5 Ga), the Sveconorwegian (1.25–0.85 Ga) and the Caledonian (0.6–0.4 Ga) events, respectively. The Svecofennian province forms the central part of the shield and is divided into northern, central and southern subprovinces (Gaal & Gorbatschev 1987; Fig. 1). Rocks of the TIB and the Gothian orogeny truncate the Svecofennian province in the southwest and the Southwest Scandinavian province, which was formed during the Gothian orogeny and heavely reworked during the Sveconorwegian orogeny. Basic dykes intruded within two different time intervals during and after the Gothian orogeny. At c. 1.65 to 1.3 Ga dykes associated with anorogenic granite emplacement intruded into the crust in the central parts of the shield. At c. 1.27 Ga dolerite sills and dykes (Welin & Lundqvist 1975; Patchett 1978; Suominen 1991), the Central Scandinavian Dolerite Group (CSDG, Gorbatschev et al. 1979), intruded in the central part of Sweden and southwestern part of Finland.

The Central Scandinavian Dolerites can be divided into five geographically separated complexes (Elming & Mattsson 2001), the Satakunta, the Jämtland, the Ulvö, the Dala and the Västerbotten complexes. It should be pointed out that by the name Ulvö complex, Elming & Mattsson (2001) include all post-Jotnian dolerites along the coastline of the Västernorrland county (Lundqvist et al. 1990), including the Ulvön island and also dolerites in the Finnish archipelago close to the city of Vasa. Flat lying basic intrusions have been identified in drilling and from seismic data in deep parts of the crust beneath the Siljan Impact Structure (Juhlin & Pedersen 1987). The sills were also interpreted from seismic data in the basement beneath the Caledonian nappes in Jämtland (Palm et al. 1991) and in the crust of the Bothnian Bay (BABEL Working Group 1993), which suggests a more or less continuous belt of dolerites from Sweden to Finland. Paleomagnetic data are consistent within and between the different complexes (e.g. Bylund & Pesonen 1987; Elming & Mattsson 2001 and references therein), which may suggest that there was no or only a small difference in time between the intrusion of the sills.

Post-Jotnian dolerites of the Ulvö complex

In the Ulvö region the post-Jotnian dolerites mainly outcrop as an oval shaped

body, with a probable lopolithic form with a radius of c. 50 km and a thickness of a few decimetres up to 500–1000 m (Lundqvist et al. 1990; Fig. 2). The dolerites of this ring shaped body mainly intruded older Paleoproterozoic basement rocks (Lundqvist et al. 1990). In the central and northern parts of the ring the sills have intruded Jotnian sandstone as well as granite, gabbro and anorthosite rocks of the Nordingrå rapakivi-type massif (Ahl et al. 1997). The geology of the Ulvö island, which is located just outside the coastline close to the northern part of the oval, is dominated by these dolerite sills and granite of the Nordingrå complex. K–Ar isochron ages are reported for the Ulvö dolerites giving an age of about 1245 Ma (Welin & Lundqvist 1975), and Rb–Sr data from the sills in central Sweden indicate an age of c. 1220 Ma (Patchett 1978). U–Pb dating of post-Jotnian dolerites in the archipelago between Sweden and Finland gives an age of c. 1270 Ma (Suominen 1991). The dolerites of the Ulvö island are thought to have been generated by several sets of intrusions, whereas the oval body belongs to one single intrusion event (Lundqvist et al. 1990) that is suggested to have taken place at a slightly deeper level than the intrusion of the Ulvö dolerites (Ahl et al. 1997). Several ENE–WSW striking post-Jotnian dolerite dykes, some more than 100 m wide, occur west of these outcrops (Lundqvist 1990) and small E–W trending post-Jotnian dolerite dykes are also reported from within the Ulvö complex. However, some of these dykes cut the sills and are thus younger (Elming & Mattsson 2001). The Ulvö dolerites are characterised by rhythmic layering indicated by the preferred orientation of tabular plagioclase (Lundqvist & Samuelsson 1973; Larson 1980) and this layering is thought to be the result of convection currents within the ancient magma chamber (Lundqvist 1990). The main minerals are plagioclase, augite, olivine, biotite and Fe-Ti oxides and the magnetisation of the dolerites is completely governed by Fe-Ti oxides (Magnusson 1983; Elming & Mattsson 2001).

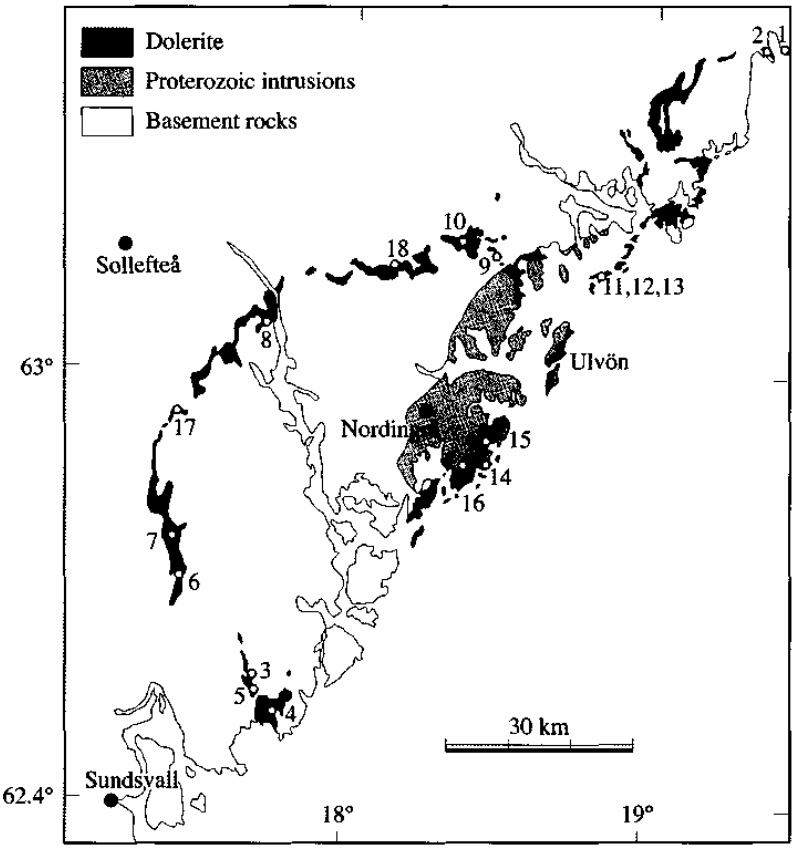


Fig. 2. Outcrop of the post-Jotnian dolerites in central-east Sweden (simplified from Piper 1979), with the sampling sites of this study. The geographic location of the area is shown in Fig. 1.

Sampling and methodology

Samples for paleomagnetic and AMS analyses were collected with a portable drill and as block samples and oriented using magnetic and sun compasses. Generally each site represents at least a 20 m^2 large outcrop. The study includes 185 samples (172 dolerite, 12 gabbro and 1 sandstone sample) collected at 18 sites distributed over an area of c. 100×50 km^2. 13 sites are situated in the outer part of the oval intrusion, three sites are situated close to the centre of the oval, where they cut rocks of the Nordingrå massif. Two sites are situated north of the oval intrusion (Fig. 2).

The specimens were demagnetised by conventional alternating field (AF) and thermal demagnetisation techniques and the remanent magnetisation was measured with a spinner (Czech-JR5) and a SQUID (2G-DC) magnetometer. Since paleomag-netic data already exist abundantly from the dolerites in the Ulvö complex (e.g. Piper 1979) only five samples were AF demagnetised and two thermally demagnetised from each site respectively. Paleomagnetism was used mainly in order to determine the post-Jotnian origin of the rocks. The different components of remanence were defined by least square fit of vectors in three-dimensional space (Torsvik 1986; Torsvik et al. 1996) and by principal component analyses (Kirschvink 1980). The vectors were generally defined with mean angular deviations smaller than 5°.

The anisotropy of magnetic susceptibility (AMS) was measured to define the magnetic fabric of the sills. In average 10 samples from each site were measured. Basically, the AMS of a rock is described by a second–order tensor (e.g. Hrouda 1982), which can be visualised by an ellipsoid with three principal axes of maximum, intermediate and minimum susceptibility (kmax, kint and kmin, respec-

tively). For multi-domain magnetite the kmax axis is parallel to the long axis of an ellipsoidal grain and for most rocks the kmax corresponds to the mineral lineation, whereas the kmin axis corresponds to the pole of the plane of magnetic foliation (Tarling & Hrouda 1993). The degree of anisotropy is characterised by P = kmax/kmin and the shape of the anisotropy ellipsoid is defined by the shape parameter T (Jelínek 1981), where:

$$T = \frac{2\ln k_{int} - \ln k_{max} - \ln k_{min}}{\ln k_{max} - \ln k_{min}}$$

For $-1<T<0$ the ellipsoid is prolate (cigar-shaped) and for $0<T<1$ the ellipsoid is oblate (disc-shaped). If T=0 the ellipsoid is neutral. The AMS and temperature variation of magnetic susceptibility of rocks were measured with a Kappabridge KLY-3 and a CS-3 instrument, respectively (Geofyzika Brno). The statistical analysis of the AMS data was performed using a technique of Jelínek (1978).

Elongated particles in a flowing magma line up with the long axis parallel to the flow direction (Knight & Walker 1988), and when the magma crystallizes the oxides will mimic the fabric of the pre-existing silicate phases (Tauxe et al. 1998). If ferromagnetic grains of multi-domain size dominate the magnetic susceptibility of the rock the preferred orientation of the kmax axis will be parallel to the flow direction of the magma. Since Knight & Walker (1988) demonstrated the use of AMS as an indicator of direction of magma flow in dykes the method has been used in several studies to define flow fabrics (e.g. Bates & Mushayandebvu 1995; Raposo 1997). Elongated particles in a flowing magma become imbricated along the chilled margin of the dyke (Knight & Walker 1988), so by measuring the AMS of samples collected within c. 10 cm from the margin it is possible to determine not only the lineation but also the true direction of the magma flow. Dragoni et al. (1996) presented a theoretical model of the behaviour of rigid ellipsoidal grains in a flowing magma driven by a pressure gradient. They compared the model results with AMS fabrics from Ferrar dolerite sills on east Antarctica and concluded that the magnetic lineation is generally a good marker of the magma flow direction, except for in the centre of the sill body that is strain free and no preferred orientation can be defined. Large scatter or absence of preferred directions in the magnetic lineations may thus indicate that the measured samples were not collected close to the sill margin. In our study we could in six cases observe the contact between the host rock and the sill (sites 11–16) and at those sites we also measured the orientation of the sill planes. In three other cases we know that the samples were collected within a few meters from the contact zone (sites 1, 2 and 6). At the other nine localities it is thus not possible to determine where in the sill body the samples were collected and the scatter of the directions are sometimes high. The sills at the sites 11, 12, 14 and 16 are 0.1–0.5 m thick and represent minor magma intrusions. On basis of statistical requirements for the confidence of mean AMS directions (Jelínek 1978) a maximum size limit for the semi axes of the 95% confidence ellipse of the mean lineation direction is 25°. Sites that have mean directions with confidence ellipses above this limit are considered too poorly defined to represent a statistically reliable direction, and are therefore rejected.

Results

Paleomagnetic results

In all18 sites it is possible to define a characteristic magnetisation. The magnetisation is usually isolated in a narrow range of unblocking temperatures of c. 550 to 600°C and in AF fields above 20 mT (Fig. 3A). There is a small portion of soft remanence, which is usually erased at 15 mT to 20 mT demagnetising fields. This

Table 1. New paleomagnetic data from the post-Jotnian dolerite sills of the Ulvö complex and from a gabbro of the Nordingrå complex.

Site	Glat/Glon	S/N	Decl.	Incl.	k	α95	Plat.	Plon.	A95
1	63.44/19.67	1/4	61	–51	46	14	–16	148	22
2	63.40/19.49	1/4	54	–30	31	17	0	149	21
3	62.56/17.69	1/5	51	–35	107	7	–1	151	10
4	62.50/17.78	1/5	53	–44	71	9	–8	151	13
5	62.53/17.72	1/5	44	–46	137	7	–7	160	9
6	62.68/17.49	1/5	47	–45	55	10	–7	157	15
7	62.73/17.45	1/5	53	–43	146	6	–7	151	9
8	63.04/17.73	1/4	33	–40	42	14	0	167	20
9	63.14/18.49	1/5	34	–52	117	7	0	168	20
10	63.17/18.35	1/5	45	–40	167	6	–3	157	8
11	63.14/18.78	1/5	47	–27	56	10	5	154	14
12	63.15/18.77	1/4	54	–45	128	8	–9	152	12
13	63.15/18.77	1/6	44	–20	12	20	9	155	24
14 Sill	62.87/18.44	1/5	38	–29	45	12	6	162	15
14 Gabbro	62.87/18.44	1/6	41	–41	177	5	–2	161	7
15	62.91/18.42	1/4	47	–32	35	16	2	154	21
16	62.87/18.33	1/4	26	–39	104	9	3	174	13
17	62.94/17.49	1/4	45	–56	46	14	–15	161	24
18	63.15/18.15	1/5	54	–40	136	7	–6	150	9
Form. mean (AF) (Dolerite sills)	62.97/18.28	18/84	46	–40	51	5	–3	157	7
Form. mean (Thermal) (Dolerite sills)	62.97/18.28	18/34	46	–42	29	5	–5	158	7

Glat/Glon = geographic latitude and longitude of the site, S/N = number of sites/samples, Decl. (Incl.) = mean declination (inclination) of the remanent magnetization, k = best estimate of the Fisher precision parameter, α95 = radius of the 95% cone of confidence, Plat (Plon) = latitude (longitude) of the virtual geomagnetic pole position, A95 = half-angle of the 95% circle of confidence of the paleomagnetic pole position. The formation mean (AF) is calculated from the site mean directions and the formation mean (Thermal) is calculated from individual samples.

soft remanence carries a magnetisation with a direction similar to that of the present earth's field (PEF). In thermal demagnetisation a low unblocking temperature component is isolated and from the similarity in direction with that of the PEF a viscous origin for this remanence is suggested (Fig. 3A). A gradual change in inclination is sometimes seen during alternating field demagnetisation. Remanence components of higher negative inclinations are isolated in higher coercivity ranges. This change in directions is probably due to overlapping coercivity spectra, between the viscous and the primary magnetisation. Such gradual changes in directions are also observed in thermal demagnetisations, but here the two components are clearly separated. The coercivity of the natural remanent magnetisation (NRM) is generally high, with median destructive fields (MDF, Dunlop & Özdemir 1997) of 26±9 mT for 16 of the sites. The average Q-value (Königsberger ratio, e.g. Collinson 1983) is Q=1.1, ranging from Q=0.1 to Q=10.9, which is in good accordance with results previously reported by Magnusson (1983). This suggests that the carrier of remanence is predominantly pseudo-single-domain magnetite-titanomagnetite. However, at two sites (12 and 16) the characteristic magnetisations have extremely high coercivities and only a small portion is erased in AF fields of 100 mT (Fig. 3B). These two sites also have significantly lower mean susceptibilities (Table 2) and at site 12 the rock has a rather high degree of AMS (P=1.15). Both sills are thin, with a thickness of 0.1–0.3 m and the host rock is red sandstone. At site 16 the basic magma has intruded as thin sheets between layers of sandstone and at site 12 parts of the sill seems to have been mixed with the sandstone. Thermal demagnetisation shows that hematite governs the NRM of these rocks completely (Fig. 3B), which is likely to explain the high coercivity as well as the higher degree of AMS. A large peak in the susceptibility versus temperature diagram at T=150–350°C for the samples from both sites indicates chemical alteration during heating, possibly maghemtization (Dunlop & Özdemir 1997), however, most of the induced magnetisation seems to be carried by magnetite and not hematite (Fig. 4D).

From the AF demagnetisations site mean directions with precision parameters k (Fisher 1953) ranging from 12 to 167 are defined (Table 1). These direc-

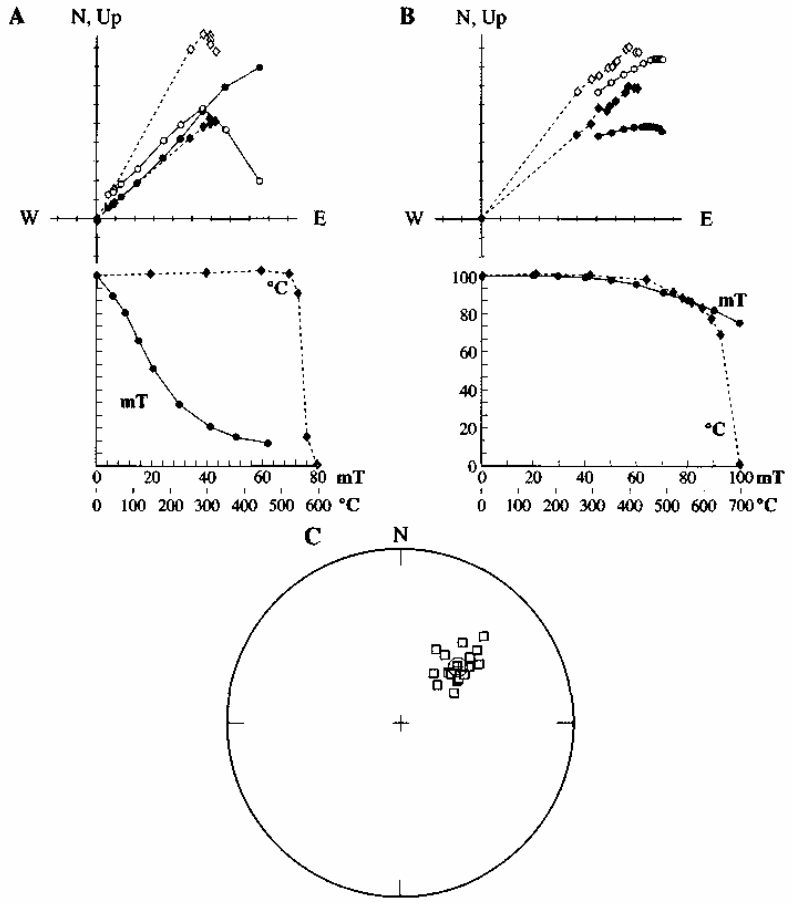


Fig. 3. **A** and **B**, examples of remanence behaviour during progressive alternating field and thermal demagnetisation of dolerite sill samples. Open (closed) symbols in the vector plots denote vertical (horizontal) projections. Diamond (ring) symbol indicates thermal (alternating field) demagnetisation respectively; C, Lower hemisphere equal area projection plot of site mean remanence directions of the dolerite sills. The cross denotes the formation mean direction and the ellipse shows the 95% cone of confidence of the formation mean direction.

Table 2. AMS parameters for the dolerite sills of the Ulvö complex

Site	N	Kmean	P	L	F	dec Kmax	inc Kmax	α95 Kmax
1	10	23400	1.03	1.01	1.02	197	10	42
2	10	17700	1.03	1.02	1.01	308*	12	15
3	10	17500	1.04	1.01	1.03	111	1	44
4	10	33400	1.07	1.01	1.06	305	9	37
5	10	17800	1.08	1.01	1.07	169	0	45
6	10	21600	1.02	1.01	1.01	173*	14	13
7	10	26300	1.04	1.02	1.02	346*	15	24
8	10	15900	1.05	1.01	1.05	146	18	60
9	10	18000	1.11	1.02	1.09	236*	12	22
10	10	36400	1.05	1.01	1.04	235	9	43
11	10	54000	1.08	1.01	1.07	94	12	35
12	9	270	1.15	1.03	1.11	148*	7	15
13	9	21800	1.03	1.01	1.03	80	9	27
14	9	1000	1.01	1.00	1.00	344*	18	17
15	6	50000	1.05	1.01	1.04	219	6	36
16	9	170	1.11	1.01	1.09	120*	7	12
17	10	22200	1.02	1.01	1.01	126	20	45
18	10	24000	1.02	1.00	1.01	319	2	55

N = number of samples, Kmean = mean susceptibility (10^{-6} SI), P = degree of anisotropy, L = degree of lineation, F = degree of foliation, dec Kmax = declination of the magnetic lineation, inc Kmax = inclination of the magnetic lineation, α95 Kmax = largest semi-axis of the confidence ellipse about the Kmax axis.
* denote statistically reliably direction used in figure 6B.

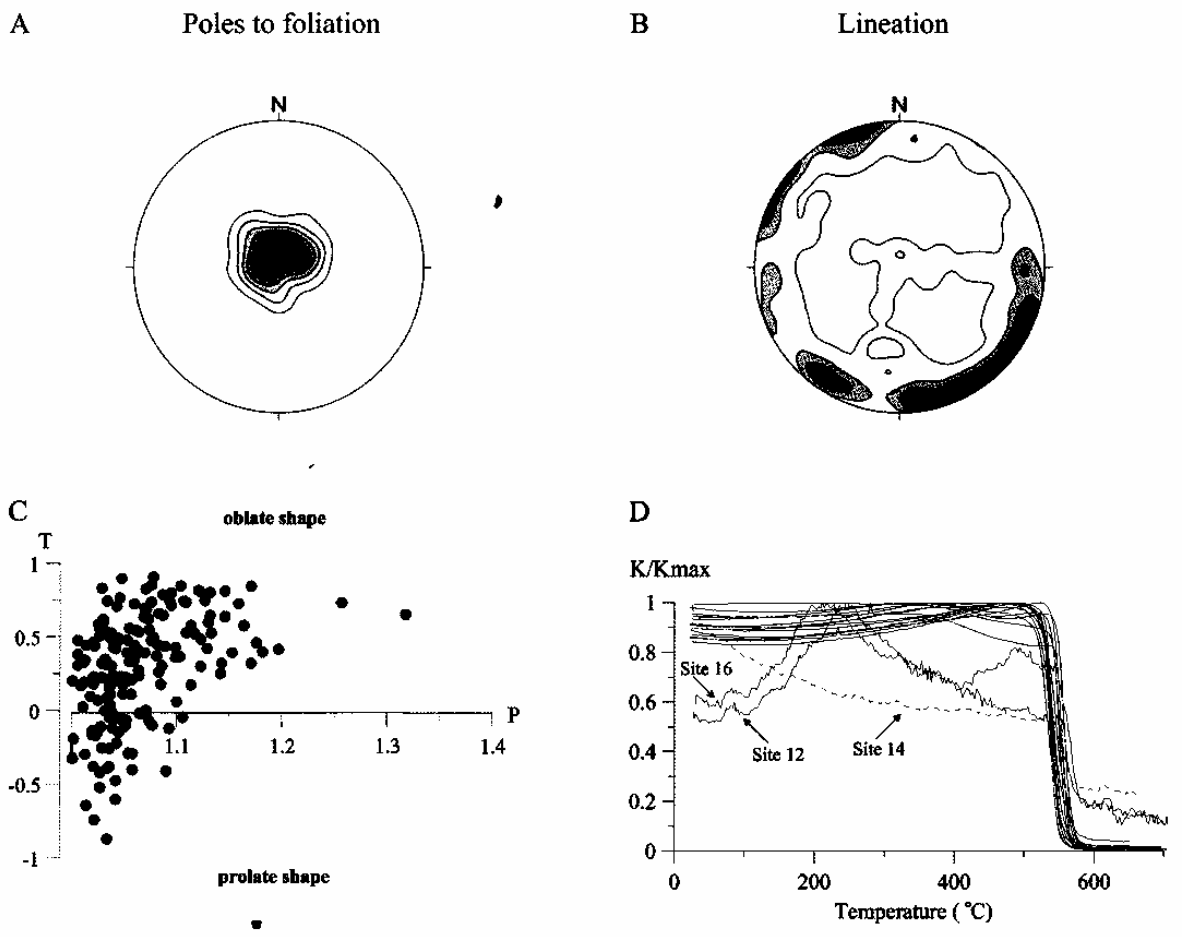


Fig. 4. AMS data of the dolerite sills. **A** and **B**, contour plots of directions of minimum (A) and maximum (B) susceptibility axes in stereographic projections of all samples (n=172); **C**, P–T diagrams (Jelínek 1981) of all samples; **D**, heating curves of the mean susceptibility of one sample from each site.

tions are distributed within a narrow range of declinations and inclinations (Fig. 3C). By combining the site mean directions, a formation mean direction is defined with decl. =45.9° and incl. =–39.9° (α95 =4.9° and k=51), from which a paleopole position of Plat=–3° and Plon=157° (dp=4° and dm=6°) is calculated. This pole position is in very good agreement with those reported from post-Jotnian sills and dykes in previous studies (e.g. Piper 1979 and references therein; Elming & Mattsson 2001 and references therein). A tilt correction of the site mean directions with reference to the orientation of the magnetic foliation plane at each site (Fig. 6A) yields no significant change of the formation mean direction (corr. decl.=42.9°, corr. incl.=–37.7°), but the α95 increases to 5.3° and the k-parameter decreases to 43. The changes of the α95 and the k indicate a deterioration of the grouping of the directions after correction. This may suggest a negative fold test for the sills (e.g. Collinson 1983), which could indicate that the characteristic magnetisation was formed when the sills were in their present position.

Alternating field demagnetisation was performed on six samples from the Nordingrå gabbro hosting the sill at site 14. The samples were collected along a profile perpendicular to the sill, starting at the contact zone and ending at 64 cm distance from the 10 cm thick sill. The intensity of magnetisation generally decreases exponentially but the coercivity of the natural remanent magnetisation (NRM) is fairly high, with an average median destructive field (MDF, Dunlop & Özdemir 1997) of 20 mT. Vector plots of the AF directions reveal one stable high coercivity components for all six samples. The resultant mean remanence direction of decl.=41° and incl.=–41° (α95=5°; Table

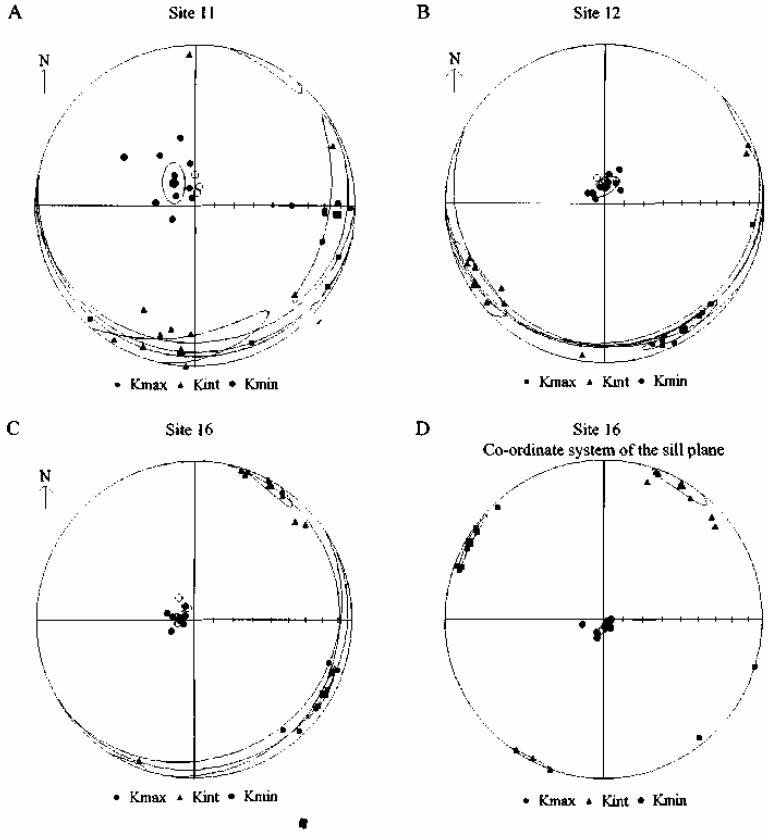


Fig. 5. **A–D**, examples of lower hemisphere equal area projection plots of AMS data from three sites where the orientation of the sill plane is known. The ellipses are 95% confidence limits about each axis of the site mean susceptibility ellipsoid. The great circles show the orientation of the plane of the sills as defined in field and the small open rings denote the corresponding poles to each sill plane, respectively. In (D) the AMS axes of site 16 (C) have been rotated with reference to the sill plane. This shows that the mean maximum axes direction is different from the sill plane, which indicates an imbrication pattern.

1) is not significantly different from the formation mean direction of the sills. These data therefore represent a negative baked contact test for the sill (McElhinny 1973), meaning that the gabbro at this site and the thin sill have been remagnetised after the intrusion of the sill.

AMS results

The magnetic fabrics of the sills are very consistent within and between the sites. The minimum principal axes are clustered with a subvertical direction (Fig. 4A) and maximum and intermediate axes are subhorizontal and they generally show a girdle distribution, but often they are also clustered (Fig. 4B). From the contour plot of Fig. 4B it is evident that the lineations mainly cluster in the NW–SE and NNW–SSE directions. The average mean susceptibility is 0.022 SI (n=172) and the anisotropy parameters averages at P=1.07±0.05, L=1.02±0.01 and F=1.05±0.04. A majority of the samples have an oblate shaped AMS ellipsoid and a degree of anisotropy (P) below 1.10 (Fig. 4C). The sites 12, 14 and 16 have significantly lower mean susceptibilities and the sites 12 and 16 also have higher degrees of anisotropy (Table 2). Susceptibility versus temperature data indicate that magnetite governs the AMS of the sills in all 18 sites, though site 14 has an obvious paramagnetic contribution to the susceptibility as well (Fig. 4D).

In the sites with exposed contacts between the sill and its host rock (sites 11–16), we find that the maximum and intermediate principal anisotropy axes always lay in, or close to, the plane of the sill (Figs. 5A–C). The minimum axes fall close to the pole of the dyke plane. This is an indication that the AMS data is primary and reflects the ancient magma flow. For seven of the eighteen sampled sites we can define a mean lineation direction with a confidence ellipse with radius smaller than 25° (Table 2). Five of the sites show NW–SE, NNW–SSE lineation directions and the other two show NE–SW, NNE–SSW directions. At site 16 the site mean lineation (kmax) is slightly but significantly different from the orientation of sill plane as measured in the field. This significant but slight difference suggests an imbrication pattern (Fig. 5D). The samples were collected from the lower contact of the sill body and the AMS fabric indicates a possible magma flow direction from the present north-west towards south-east.

The lopolithic shape of the oval intrusion seems to be confirmed in this study by the dip directions of the magnetic foliation planes (Fig. 6A). At 8 of the 12 sites located in the ring shaped outcrops (including 2 sites in the Ulvö island, Mattsson & Elming 2001) the planes roughly dip toward the centre of the oval shaped structure, which is located in the vicinity of the sites 14–16.

Discussion and conclusions

The purpose of the paleomagnetic measurements in this study is mainly to try to guarantee that the dolerite sills we use for our magma flow interpretation are of post-Jotnian age (c. 1270 Ma). Piper (1979) presented paleomagnetic data from 43 sites of the Ulvö comlex that yield a paleomagnetic pole position of Plat=–7.5°, Plon=156.5°. All our samples have the same polarity and the directions of magnetisation correlate well with results from other studies in this area (e.g. Porter 1976; Magnusson & Larson 1977) and also from other parts of the CSDG complex (e.g. Neuvonen 1965; Neuvonen 1966; Bylund 1985; Elming & Mattsson

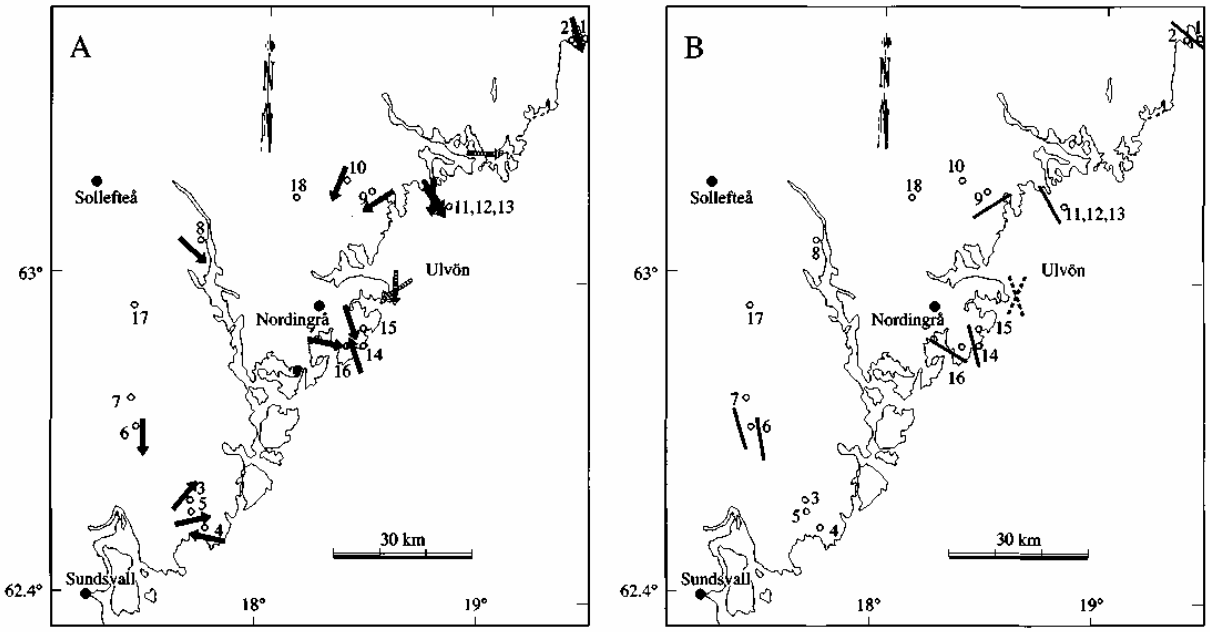


Fig. 6. **A**, Arrows showing the site mean dip direction of the magnetic foliation planes of the dolerite sills with reference to the geographical distribution of the sites. The three hatched arrows are redrawn from Elming & Mattsson (2001); **B**, Site mean strike directions of the magnetic lineation from 9 sites (7 sites from this study (Table 2) and 2 sites from Elming & Mattsson (2001), denoted by hatched lines). The dip of the lineation is most often close to zero and is not accounted for in the figure.

2001; Table 3). In Fig. 7 the mean paleomagnetic pole position calculated from the data of this study is plotted on a segment of the apparent polar wander path (APWP, Elming et al. 1993) for the Fennoscandian Shield together with some of the pole positions reported in previous studies of dolerite sills from this area, and also from other parts of the shield. As seen in the figure there is generally no significant difference between our pole position and those previously reported for the CSDG complex.

The negative baked contact test reported for the sill at site 14 (Fig. 2) is a possible argument against a post-Jotnian origin of this rock. The gabbro hosting the sill belongs to the Nordingrå rapakivi-type massif, which is dated at c. 1.58 Ga (U–Pb, Andersson 1997 and references therein). Paleomagnetic pole positions (Plat=33–37° and Plon=138–149°, Piper 1980) reported from granite and gabbro-anorthosite rocks of the Nordingrå massif fall close to the 1500 Ma reference pole of the APWP for the Fennoscandian Shield (Elming et al. 1993). The paleomagnetic pole position (Plat=–2° and Plon=161°) determined in this study for the gabbro at site 14 indicates an age of magnetisation of c. 1.27 Ga. Based on our paleomagnetic data only, it is possible that the sill does not belong to the post-Jotnian intrusions and that it intruded in the gabbro sometime after 1.58 Ga, prior to 1.27 Ga. After that the post-Jotnian igneous event could have given rise to a remagnetisation of both rocks. A regional remagnetisation can be excluded since the pole positions of the Nordingrå massif reported by Piper (1980) (see above) are significantly different from the pole position of the gabbro at site 14. If this sill is not of post-Jotnian age, the most probable igneous activity that could

Table 3. Previous paleomagnetic data of post-Jotnian dolerites

Dyke swarm	Glat/Glon	S/N	Decl.	Incl.	k	α95	Plat.	Plon.	A95	Ref.
Mean Ulvö dolerites	63.6/18.5	2/52	45	–37			–2	157		1,2
Vasa dolerite dykes	63.0/20.9	15/108	38	–29	54	5	7	164	4	3
V. Norrland dolerites	62.9/18.3	43/252	48	–49	71	3	–8	157	4	4
Mean Nordingrå dolerites	62.8/18.4	2/21	38	–38			0	163		5,6
Gnarp sills	62.1/17.3	2/12	60	–48			–13	146		7
Gnarp dolerites	62.1/17.2	1/6	46	–52	78	8	–11	159	8	7
Ulvö Complex	63.0/18.4	4/37	48	–45	77	11	–8	157	11	8

The mean directions are calculated from the site means when data from more than one site exist. For other explanations see Table 1. 1 = Magnusson & Larson (1977), 2 = Larson & Magnusson (1976), 3 = Neuvonen (1966), 4 = Piper (1979), 5 = Fridh (1979), 6 = Piper (1980), 7 = Poorter (1975), 8 = Elming & Mattsson (2001).

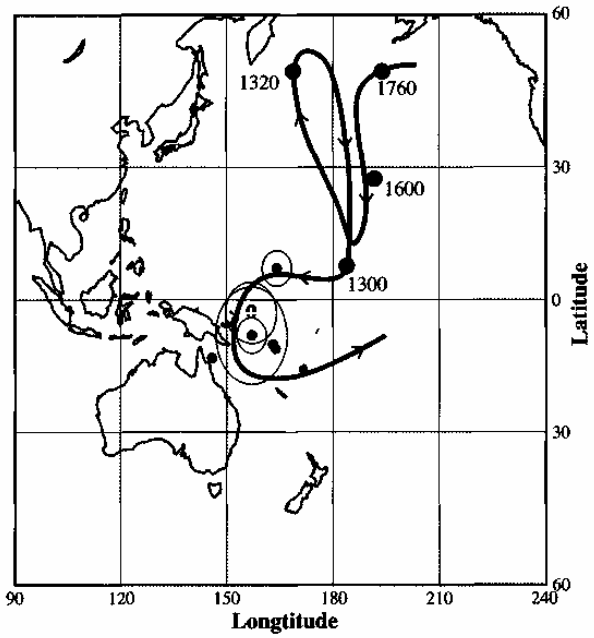


Fig. 7. Mean virtual geomagnetic pole (VGP) positions plotted together with A95, the half-angle of the 95% circle of confidence. Black dots indicate old data from different parts of the CSDG (see Table 3) and the white cross denotes the VGP position calculated from this study.



Fig. 8. Tectonic reconstruction of Laurentia and Baltica at 1.27 Ga. The arrows denote the flow direction of magma in the sills in Fennoscandia (full drawn line) and Greenland (hatched line) as indicated from AMS and geological data. The picture is re-drawn from Mattsson and Elming (2001).

have generated the sill is the emplacement of the Ragunda massif, which is c. 70 m.y. younger than, and situated west of the Nordingrå massif. However, the only dykes that are reported from the Ragunda massif have a granitic composition and they have an east–west strike with a vertical dip (Ahl et al. 1997). The most probable explanation to the negative baked contact test is that the sill at site 14, which is only c. 0.1 m thick, is a sister intrusion to a larger main sill body that remagnetised large parts of the gabbro (possibly also the sill), and now is eroded away. A large sill body outcrops in several places only a few kilometres from the site 14, e.g. at site 15 (Fig. 2). We therefore suggest that all 18 dolerite sills sampled in this study most likely are of post-Jotnian age.

The degree of AMS (P) is below 1.10 for a majority of the samples, which is the usual for primary igneous rocks (Hrouda 1982). The sill at site 12 has a mean anisotropy degree of P=1.15 and an unusually low mean susceptibility of K=0.27 mSI. According to the thermal demagnetisations, hematite completely governs the remanent magnetisation of this rock so it is possible that the hematite also affects the AMS. However, magnetite seems to be the dominant carrier of the magnetic susceptibility at all 18 sites (Fig. 4D). Hematite also dominates the remanent magnetisation at site 16 but still the mean degree of anisotropy is only P=1.11. Visual inspection of the rock samples and outcrops, does not show any evidence of chemical alteration, as e.g. chlorite alteration. Since both sills intrude in the Jotnian sandstone they are clearly post-Jotnian. What we do find, in these two sites only, is that the rock has a lighter colour possibly indicating that the basic magma was mixed with the sandstone during the intrusion of the sills. According to Magnusson (1983) hematite occurs both as rounded particles and as lamellae in crystals in the Jotnian sandstone. Thermal demagnetisation of a sandstone sample collected at site 12 confirms the presence of hematite in this rock. The sample was collected close to the contact zone and its remanence direction is parallel to the sill direction. The sandstone was therefore presumably re-heated during the intrusion of the sill, which may have led to the formation of secondary hematite. However, together with Magnusson's (1983) observations it seems possible that the hematite in the sills at the sites 12 and 16 originate from the sandstone.

At the sites with exposed contacts between the sill and its host rock (sites 11–16), we find that the maximum and intermediate principal anisotropy axes always lay in, or close to, the plane of the sill. This is an indication that the AMS fabrics are primary and reflects the ancient magmatic flow directions during the emplacement of these rocks. The AMS data indicate fairly uniform NW–SE directed lineations of the dolerite sills, also when two sites from the Ulvö island are included (Elming & Mattsson 2001; Fig. 6B). Two of the nine sites have lineation directions that deviate from the NW–SE pattern, both trending c. NE–SW. Six of the sites with well defined lineations belong to the main flow whereas three sites (12, 14 and 16) are small sills that prob-

ably constituted sister intrusions to the main flow. We interpret these lineation directions as indicators of a preferred NW or SE directed magma flow at the time of the intrusion of the sills. The flow direction does not seem to be correlated to the oval shape of the intrusion. The preferred NW–SE lineation directions fit nicely to the c. NW or SE directed magma flow of the CSDG-dolerites in northern and central Sweden shown by Elming & Mattsson (2001). The imbrication pattern defined at site 14 suggests that the magma propagated from the present NW towards SE.

In a plate tectonic reconstruction model based on paleomagnetic data and geological criteria it is proposed that Baltica and Laurentia were joined together c. 1.27 Ga, with NE Greenland attached to NW Baltica (Buchan et al. 2000). AMS measurements of dolerite sills from the CSDG from northern and central Sweden indicate a fairly uniform NW or SE directed magma flow at that time (Elming & Mattsson 2001). Corresponding measurements on dolerites and basalts in NE Greenland, as well as geological evidence, indicate a SW flow direction (Marcussen & Abrahamsen 1983), which in the tectonic reconstruction is parallel to the flow of the sills in Sweden (Fig. 8). Elming & Mattsson (2001) proposed that Central Scandinavian Dolerite Group is an expression of the break up of Baltica from Laurentia some 1.27 Ga ago. The new AMS data from the Ulvö complex presented in this study fit well in to the magnetic fabric pattern shown by Elming & Mattsson (2001) and support their plate tectonic model.

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